

SkillNest — Server Configuration Roadmap

Everything to configure on the server for Phase 1 (mostly already required for production) ? Phase 2, with the **reason** each item exists. Follow in order to avoid missed steps.

Phase 1 — production hardening (do before real users)

1. PHP 8.2 runtime (requires compiled extensions)

EXTENSION	WHY
<code>pdo_mysql</code> , <code>mysqlnd</code>	Database access (P1 uses MySQL).
<code>bcmath</code>	Currency math (orders, fees, wallet, payments) — avoids float rounding.
<code>gd</code> (or <code>imagick</code>)	Image uploads/resizing (avatars, service images, KYC doc previews).
<code>fileinfo</code> , <code>mbstring</code> , <code>xml</code> , <code>curl</code> , <code>openssl</code> , <code>zip</code> , <code>ctype</code> , <code>tokenizer</code>	Required by Laravel core + DomPDF (invoices, privacy PDF), file uploads, PDF/image libs, HTTP outbound (AI/payment/webhook later).
<code>intl</code>	Laravel number/date formatting; needed by framework validators on some ops.
<code>opcache</code> (enable in prod)	~2–3× PHP speed; critical for any traffic. <code>opcache.enable=1</code> , memory ? 128M, <code>validate_timestamps=0</code> when code is deployed cleanly.
(P2) <code>redis</code> ext.	Queue + cache + session + webhooks (P1 keeps queue=sync / cache=file but that must change in P2).

2. Nginx

- Serve Laravel via PHP-FPM (root ? `public/` , try_files to `index.php`).
- **Static asset caching** for `public/build` (Vite hashed files) — cache headers via `immutable` ; this is where the 4 apps (main/admin/seller/buyer) get fast.
- Gzip/Brotli on JS/CSS/JSON; proper `expires` on images/docs thumbnails.
- `client_max_body_size 25m` + matching `upload_max_filesize` / `post_max_size` in php-fpm — buyers upload order requirements docs, sellers upload KYC (Aadhaar/PAN) and gallery images.
- Redirect `www?apex`, `HTTP?HTTPS`.

3. HTTPS (Let's Encrypt)

Payments (CCAvenue), login, KYC docs, OTP, webhooks all need TLS. Install `certbot` , auto-renew cron, HSTS + secure headers (CSP, X-Frame-Options, Referrer-Policy) at Nginx level.

4. MySQL tuning

- MariaDB/MySQL 8.x; UTF8MB4 default (Hindi/local names), `innodb_buffer_pool_size` ? 50–70% RAM .
- Slow query log during load; index hot tables (orders.status, orders.payment_status, services.status/category, users.role).
- Backups: nightly `mysqldump` /XtraBackup to off-box storage + test-restore (money data!).

5. Laravel prod bootstrapping

- `APP_ENV=production` , `APP_DEBUG=false` , real `APP_KEY` , fresh `APP_URL` .
- `php artisan storage:link` — public images (avatars, service images, order docs, KYC uploads stored on the `public` disk are referenced via `/storage/...`). *Required or uploads 404.*

- Cache: `php artisan config:cache` , `route:cache` , `view:cache` after each deploy (currently config/route NOT cached).
- Migration on deploy: `php artisan migrate --force` (NOT migrate:fresh on live).

6. Scheduler cron

```
* * * * * cd /var/www/skillnest && php artisan schedule:run >> /dev/null 2>&1
```

P1 already has time-based logic needs (auto-accept timers, invoice numbering safety, KYC reminders in P2). Even with zero tasks today, install the cron now so future tasks just work.

7. SMTP (real email)

Admin currently configures SMTP inside the app (MailService + db settings). Production needs: real SMTP provider (SendGrid/Mailgun/Postmark or transactional SMTP), DNS SPF+DKIM+DMARC so welcome/KYC/OTP mails don't land in spam.

8. Security basics

- Fail2ban on SSH/payment endpoints; firewall only 80/443/22; disable root login.
- Deploy user owns files (no world-writable storage/session folders); `.env` out of web root.
- Session/queue/lock perms — `storage` writable by `www-data` only.

Phase 2 — scale infrastructure (when pushing webhooks/microservices/AI)

9. Redis server (+ php ext)

Reliable async so DO webhooks/AI/SMS don't block users. Set `QUEUE_CONNECTION=redis` , `CACHE_STORE=redis` , `SESSION_DRIVER=redis` . Add AUTH + memory allocation, persistence (AOF) if cache-only can use RDB.

10. Queue workers (Supervisor)

Supervisor config for `queue:work redis` (workers=countCPU*2; restarts on crash), plus separate workers for `webhooks` / `ai` / `notifications` queues to isolate load. Optional Horizon dashboard (L9-compatible): `horizon:up` .

11. Webhook delivery daemon + retries

Outgoing webhooks get their own queue + retry/backoff + signature (HMAC-SHA256 of payload) + `webhook_deliveries` table. DoS-safe incoming endpoints with secret validation + idempotency keys.

12. Search service (Meilisearch / OpenSearch)

Run on same box (small) or separate VPS; seed from services/sellers via jobs; geo filters for local "near me" hits. Search is "nice in P2, required for the local marketplace feel".

13. Object storage / CDN

Move uploads to MinIO/S3-compatible (KYC docs, order docs, gig media) with CDN (Cloudflare/Bunny) for images. Keep `public` disk fallback for avatars if cost matters.

14. AI gateway host

Where ai-service lives (FastAPI, Python 3.10+, GPU box optional for local LLaMA/Whisper). Secure: API-key auth, queue-delivered jobs, separate from main app to avoid LLM call latency blocking web.

15. Notification channel providers

- SMS gateway credentials (MSG91/Fast2SMS) — env secrets.
- WhatsApp Business API (Gupshup/360dialog) — webhook URL internal + IPN.
- SMTP stays from #7.

India-first delivery: email alone insufficient for local/non-technical sellers; notifications are the #1 retention lever.

16. Monitoring / observability

- Error tracking: Sentry or Flare (Laravel SDK).
- Health: Laravel PULSE equivalent for L9 (spatie/laravel-health / cache + queue dashboards); UptimeRobot for public + webhook endpoint URLs.
- **Logs**: JSON logs with request_id, shipped to Loki/ELK or S3 for later analysis (needed to debug webhook chains + AI usage).

17. CI/CD + deployments

GitHub Actions: `php-cs/lint`, `php artisan test` (or at least `route:list` + migrate check), `npm build` (4 apps) + `fix-manifest`, then deploy via SSH/rsync or Docker image. Deploy script runs: `maintenance-mode ? git pull ? composer install --no-dev --optimize-autoloader ? migrate --force ? config/route/view cache ? queue:restart ? up`.

18. Optional: gateway / message broker for microservices

If services split wide: event bus = Redis Streams or RabbitMQ, plus API gateway (Nginx or Laravel routes forwarding to service domains). Only needed when team >1 / true deployment independence; the roadmap assumes events-first monolith ? extract services one at a time.

19. Environment secrets inventory (new in P2)

SECRET	USED BY
REDIS_HOST/AUTH	queue/cache/session
OPENAI (or ANTHROPIC/GEMINI) key	ai-service LLM
MEILI_MASTER_KEY / OpenSearch creds	search-service
MINIO (or AWS) keys	object storage
MSG91 / Fast2SMS keys	SMS
WABA phone_number_id + token	WhatsApp
CCAvenue keys already (P1)	payments
DIGILOCKER (client id/secret)	eKYC (P2)
Sentry DSN	errors

Order to follow: 1?8 (Phase 1 go-live), then 9?19 for Phase 2. If you use a managed DB / managed Redis (Aiven, PlanetScale, Upstash) many of the tuning items become config-only — still keep #5, #7, #8.